

Determination of Orange Maturity Level Using Probabilistic and IBL Techniques

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Abstract. Several characteristics, such as quality, color, texture, etc. should be considered to determine the price of vegetables and fruits. Generally, they are classified manually and usually some errors in the product categorization occur due to the subjectivity of persons with limited skills and long hours' work. Computer vision systems can be used to help in classification tasks. These systems capture the image of the product and detect its relevant characteristics that define which class is the most appropriate. Therefore, these systems need some type of intelligence in order to work correctly. Artificial intelligence techniques provide several methods that can be implemented. In this article image descriptors with Naïve Bayes and K-Nearest Neighbor algorithms are used to classify orange images taking into account their level of maturity.

Keywords: Computer Vision Systems, Naive Bayes, K-Nearest Neighbor, product classification, image descriptors, IBL.

1 Introduction

In order to satisfy consumer demands, farmers classify their products in a processing line and the selection is made by human beings. However, human perception can be easily misled. Apart from that, experienced inspectors are not only difficult to find, but also expensive to pay for. Moreover, it is extremely difficult to standardize the classification results. Consequently, it is necessary to establish objective selection criteria.

The visual properties of fresh vegetable products result in very important criteria considered by consumers during the selection at the market. Also, these properties have great relevance in price definition. A visual impression of the product is created by the combination of color, size, shape and texture [9].

Computer vision systems (CVS) are used for improving the classification process. Camera-computer systems consist of light source, digital cameras, computer and image management software. CVS application examples can include evaluation of pork meat color, cereal grains classification, measurement and classification of corn whiteness, automated sorting of pistachios, inspection of Golden Delicious apples,

detection of pinhole damage in almonds and bone detection in fish and chickens [2], [7], [8], [15].

Several applications for the inspection and classification of food quality and modeling have been successfully developed using artificial intelligence techniques as clustering, fuzzy logic and artificial neural networks (ANNs).

Classification tasks can be made for two types of learning: (a) determining the assignation rules for each observing the class to which it belongs or (b) determining if there are classes in which these observations can be grouped. The first type is known as supervised learning and the second one as unsupervised learning.

In this article, two supervised learning classification algorithms based on instances for orange image classification are presented. In section 2 some image descriptors used in classification process are mentioned. Two classification methods are explained in section 3. In section 4 the proposed model is described. The simulations and results are showed in section 5. Finally, conclusions and future work are enunciated.

2 Image Descriptors

Images are made up of features that describe them, such as color, texture, pixel, etc. For their representation, image descriptors are used. They allow to emphasize some elementary characteristics and transform all pixel set into a quantitative representation that identifies them [4].

Descriptors can be used to characterize complete images, their divisions or shades, such as Red-Green-Blue (RGB) tones [11], [12].

By using the descriptors in fragmented images for some property, greater representation is achieved. It permits better differentiation between the different images. Image descriptors commonly used are listed below [4]:

- **Median (M):** The value that divides an ordered set of observations about the scale of values, so that the number of data above the median is equal to the number of data below it.

$$\bar{x} = \begin{cases} X_{(\frac{n+1}{2})} & \text{if } n \text{ is odd number} \\ \frac{X_{(\frac{n}{2})} + X_{(\frac{n}{2})+1}}{2} & \text{if } n \text{ is even number} \end{cases} \quad (1)$$

where $\bar{x}$ = median, X = observations, n = all observations

- **Arithmetic Mean (AM):** Represents the physical center of the data set and defines the sum of the observed values divided by the number of observations.

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \quad (2)$$

where $\bar{X}$ = mean, X_i = observations, n = all observations

- **Kurtosis Coefficient (K):** A number whose magnitude indicates if data are symmetrically distributed in normal way, steeper or flatter than normal curve.

$$k = \frac{\sum_{i=1}^k f_i(x_i - \bar{x})^4 / n}{s^4} \quad (3)$$

where $k = \text{kurtosis}$, $s = \text{standard deviation}$, $x_i = \text{observations}$, $\bar{x} = \text{mean}$

— **Correlation coefficient (C):** It determines the force intensity with which two variables X, Y are linearly related, it is the Pearson correlation coefficient.

$$r = \frac{\sum_{i=1}^k (X_i - \bar{X})(Y_i - \bar{Y}) / n}{S_x S_y} \quad (4)$$

where $r = \text{correlation}$, $X_i = \text{observations of } X$ and $Y_i = \text{observations of } Y$

$S_x = \sqrt{\frac{\sum (X_i - \bar{X})^2}{n-2}}$, $S_y = \sqrt{\frac{\sum (Y_i - \bar{Y})^2}{n-2}}$, $\bar{X}$ and $\bar{Y}$ are mean of X and Y respectively

— **Variance (V):** Represents the product of the squared deviations.

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n} \quad (5)$$

where $s^2 = \text{variance}$, $x_i = \text{observations}$, $\bar{x} = \text{mean}$

— **Standard Deviation (D):** Represents square root of variance.

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}} \quad (6)$$

where $s = \text{standard deviation}$, $x_i = \text{observations}$, $\bar{x} = \text{mean}$

It is important to mention that image descriptors calculation variables are expressed in 2D arrays [11].

3 Learning Methods

In machine learning, Instance Based Learning (IBL) is a family of learning algorithms that generates classification predictions using only specific instances and its learning method consists of simply storing the presented training data. Then when a new instance is presented, a set of similar related instances is retrieved from memory and used to classify the new query instance. K Nearest Neighbor (KNN) [14] method belongs to this type of algorithms.

IBLs are sometimes referred to as “lazy” learning methods because they delay processing until new instance must be classified. A key advantage of this kind of delayed, or lazy, learning is that instead of estimating the target function once for the entire instance space, these methods can estimate it locally and differently for each new instance to be classified. Storing and using specific instances improves the performance of several supervised learning algorithms [1], [3], [13], [14].

The other hand, bayesian learning algorithms calculate the explicit probability for hypotheses, such as Naive Bayes (NB) classifier, which is a simple probabilistic

classifier based on applying Bayes' theorem and its learning method provides a probabilistic approach to inference.

In this paper both, KNN and NB methods are used for orange classification and they are tested.

3.1 Naive Bayes

The NB classifier [10], [14], [16] applies to learning tasks where each instance x is described by a conjunction of attribute values and where the target function $f(x)$ can take on any value from some finite set V .

Given a sample x represented by k values, the NB classifier is based on finding the most probable hypothesis to describe at this example. If the description of this example is given by the values $\langle a_1, a_2, \dots, a_n \rangle$ the most probable hypothesis is that described by the instance:

$$v_{MAP} = \operatorname{argmax}_{v_j \in V} P(v_j | a_1, a_2, \dots, a_n) \quad (7)$$

We can use the Bayes Theorem to rewrite this expression as

$$v_{MAP} = \operatorname{argmax}_{v_j \in V} \frac{P(v_j)P(a_1, a_2, \dots, a_n | v_j)}{P(v_j | a_1, a_2, \dots, a_n)} \quad (8)$$

Finally the resultant expression is:

$$v_{MAP} = \operatorname{argmax}_{v_j \in V} P(a_1, a_2, \dots, a_n | v_j)P(v_j) \quad (9)$$

It can estimate $P(v_j)$ counting occurrences of v_j example in the training set and divide it by the total number of examples that make this set. To estimate the number of times of instance x for each category $P(a_1, a_2, \dots, a_n | v_j)$, the whole training set must be taken into account.

The NB classifier is based on the simplifying assumption that the attribute values are conditionally independent given the target value. In other words, the assumption is that given the target value of the instance, the probability of observing the conjunction $a_1, a_2 \dots a_n$ is just the product of the probabilities for the individual attributes: $P(a_1, a_2 \dots a_n | v_j) = \prod_i P(a_i | v_j)$. Substituting this into Equation (9), it has the approach used by the NB classifier:

$$v_{NB} = \operatorname{argmax}_{v_j \in V} P(v_j) \prod_i P(a_i | v_j) \quad (10)$$

where v_{NB} denotes the target value output by the Naive Bayes classifier. Notice that in a NB classifier the number of distinct $P(a_i | v_j)$ terms that must be estimated from the training data is just the number of distinct attribute values times the number of distinct target values –a much smaller number than if we were to estimate the $P(a_1, a_2 \dots a_n | v_j)$ terms as first contemplated.

3.2 K-Nearest Neighbor

KNN is a type of instance-based learning, or lazy learning and its algorithm is a method for classifying objects based on closest training examples in the feature space [5], [6], [14].

KNN uses a distance function to determine which set of training pattern is closer to the pattern to be classified; then the nearest pattern class will be assigned to the class of unknown pattern. Its mean, unknown pattern is assigned to the most common class amongst its k nearest neighbors, and then the object is simply assigned to the class of its nearest neighbor.

KNN algorithm assumes that all instances of the class patterns correspond to points in n -dimensional space $\mathfrak{R}^n$. The nearest neighbors of an instance are defined in terms of Euclidean distance. More precisely, let an arbitrary instance x be described by the feature vector

$$\langle a_1(x), a_2(x), \dots, a_n(x) \rangle$$

where $a_r(x)$ denotes the value of the r th attribute of instance x . Then the distance between two instances x_i and x_j is defined to be $d(x_i, x_j)$, where

$$d(x_i, x_j) = \sqrt{\sum_{r=1}^n (a_r(x_i) - a_r(x_j))^2} \quad (11)$$

In KNN learning the target function may be either discrete-valued or real-valued. Let it consider learning discrete-valued target functions of the form: $f: \mathfrak{R}^n \rightarrow V$, where V the finite set is $(v_1, \dots, v_s)$.

4 Proposed Model

Using NB and KNN techniques described previously, the proposed model has the purpose of classifying images of navel oranges. The images were captured from one hundred oranges using a digital camera and its size was 221x221 pixels. The examples showed different maturity levels and they were visually classified into five classes according to their color. The image descriptors which were used were Mean (M), Kurtosis Coefficient (K) and Correlation Coefficient (C) due to the fact that they showed better categorization of examples. Therefore, MKC descriptors for each image (according to each RGB tone) were extracted and the training patterns were built as (MKC descriptors; class). This pattern set was divided into training set (used for learning process) and test set (used for determining the network generalization level).

Both classification techniques use the same input and return the obtained winner class, but their categorization processes are different. For this case, NB (Algorithm 1) uses Gaussian distribution for classification and the training process involves determining its parameters from training set. In this case, the necessary parameters for probabilities calculation are mean and standard deviation from every class (lines 7 and 8). Additionally, the prior class probabilities should be calculated (line 6). Then, when it comes a new instance to classify, the prior class and conditional probabilities

are calculated for every class (lines 12-15). Finally, the class that has the highest probability value is the winner (line 16).

Algorithm 1. NB Classification

```
1. function naiveBayes()
2.   load trainInput, trainTarget;
3.   load testInput, testTarget;
4.   /// NB training
5.   for each class in trainInput do
6.     priorMatrix = priorProbCalculation(trainInput, trainTarget);
7.     meanMatrix = meanCalculation(trainInput, trainTarget);
8.     devMatrix = stddev(trainInput, trainTarget);
9.   end for each
10.  /// NB clustering
11.  for each example of testInput do
12.    for each class do
13.      condMatrix = condCalc(meanMatrix, devMatrix, actualExample);
14.      probMatrix = priorMatrix * condMatrix;
15.    end for each
16.    winnerClass = max(probMatrix);
17.  end for each
18. end function
```

For KNN method (Algorithm 2) previous training process is not necessary.

Algorithm 2. KNN Classification

```
1. function kNearestNeighbor()
2.   load trainInput, trainTarget;
3.   load testInput, testTarget;
4.   /// KNN clustering
5.   for each example of testInput do
6.     for each class do
7.       distanceMatrix = distanceCalculation(trainInput, actualExample);
8.       distanceMatrix = inverse(distanceMatrix);
9.     end for each
10.    winnerClass = max(distanceMatrix);
11.  end for each
12. end function
```

In this case the new instances are classified as they appear (lines 6-10) and the distances between new instance and all examples of training set are calculated (line 7). In order to avoid establishing an optimal k value, all distances are considered. To do this, the inverse of all distances are calculated and summed up (line 8). This

procedure transforms KNN classifier into a global classification method [14]. Finally, the winner class is assigned to highest distance value.

The complete model schema can be viewed in Figure 1. The entire image recognition process consists of: (a) image capture, (b) MKC descriptors extraction and (c) class determination using the NB and KNN methods.

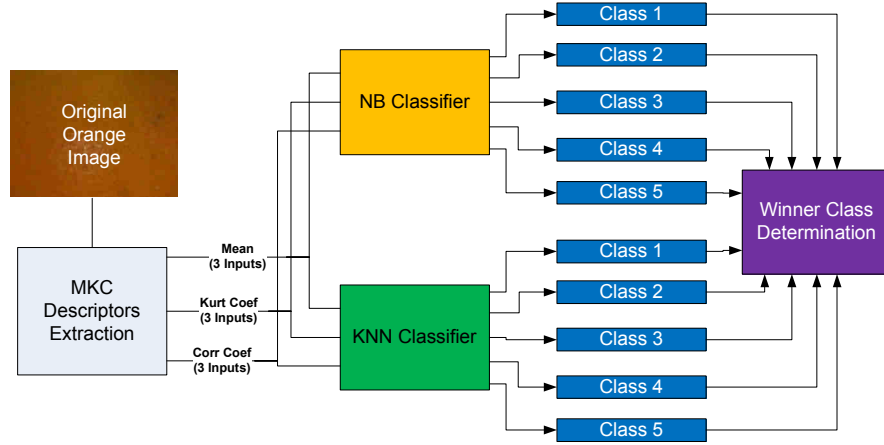


Fig. 1. Process Description

5 Simulation and Results

In order to compare the described models in previous section, the simulation was made using a set of images of one hundred oranges. For simulations training (seventy images) and test (thirty images) sets were randomly extracted from pattern set. This was made three times in order to test the NB and KNN models with different training sets. After training, descriptors for each image of the test set are presented to both mentioned models and the corresponding outputs are calculated.

For NB method, mean and standard deviation is calculated using the training set. Then the probability that each image of test set belongs to every class is calculated using the formulae explained in section 3.1. The highest value obtained corresponds to the winner class. Value “1” is assigned to winner class and “0” to the rest.

In KNN classification method, the distance to every training example is calculated for each image on test set. Then, instead of estimating an optimal k value for classification, the inverse of mentioned distance is summed up for each class. Doing this, all examples are considered neighbors but the ones which have more importance are only those nearest to the new instance to classify.

Same assignation of winner class value used in NB process is applied for KNN classification, i.e. value “1” is assigned for winner class and “0” for the rest. This procedure is applied as was explained in section 3.2.

Table 1. Simulation 1.

Target of Test Set 1																															
Image	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
Class 1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0	1	1	0	0	
Class 2	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Class 3	0	0	0	1	0	0	1	0	0	0	0	0	1	1	0	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	
Class 4	1	1	1	0	0	0	0	1	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	
Class 5	0	0	0	0	1	0	0	0	1	1	0	0	0	0	1	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	
NB Classification over Test Set 1																															
Class 1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	0	0
Class 2	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 3	0	0	0	1	0	0	1	0	0	0	0	0	0	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
Class 4	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 5	0	1	0	0	1	0	0	0	1	1	0	0	0	0	1	0	1	1	1	0	0	0	0	1	1	1	0	0	1	1	
KNN Classification over Test Set 1																															
Class 1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	0	1	0	0	0	0	1	1	0	0
Class 2	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 3	0	0	0	1	0	0	1	0	0	0	0	0	0	1	1	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0
Class 4	1	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 5	0	1	0	0	1	0	0	0	1	1	0	0	0	0	1	0	1	1	1	0	0	0	0	1	1	1	0	0	1	1	

The results are organized in tables for each method and they are then compared to the target outputs. In Table 1, for example, the images in test set 1 are written as columns and the rows are the classes (i.e. images 1, 2 and 3 correspond to class 4, image 4 corresponds to class 3, etc). Same methodology is used for output method; in these cases the mistakes in classification process are highlighted.

Tables 1, 2 and 3 have three parts: (a) the target of used test set; (b) the obtained classification of test set images using NB method and (c) the obtained classification of test set images using KNN method.

Table 1 shows the first simulation. NB and KNN classifications obtain the same results. They have a 93.33% of accuracy in class assignation over test set. Both methods fail in classification of images 2 and 17. These images correspond to class 4 and both methods mentioned classify them as class 5.

Table 2 shows the same tendency in classification process over second test set. NB obtains 90.00% of correctness and KNN decreases to 86.66%. Again, both methods fail in classification of same images (18, 21 and 26) and KNN fails also over image 27.

In third simulation (Table 3) NB method shows a non error classification while KNN fails over images 27 and 29 (93.33% of accuracy).

Taking all simulations into account, the correct classification average is 94.44% for NB method and 91.10% for KNN. An interesting point is that both methods return similar classifications, even in those cases that they fail. In these wrong classifications the winner class generally corresponds to adjacent classes. For example, in Table 1, NB and KNN methods fail in the classification of images 2 and 17 (both belonging to class 4) and they are recognized as class 5 by the two methods.

Table 2. Simulation 2.

Target of Test Set 1																															
Image	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
Class 1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	
Class 2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Class 3	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	1	0	0	
Class 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	0	0	1	1	0	0	
Class 5	0	1	0	0	1	1	1	0	1	1	1	1	0	0	0	0	1	0	1	0	0	1	1	0	1	0	0	0	0	0	
NB Classification over Test Set 1																															
Class 1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	1	
Class 2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Class 3	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	0	1	0	1	0	1	0	
Class 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	
Class 5	0	1	0	0	1	1	1	0	1	1	1	1	0	0	0	0	1	1	1	0	1	1	1	0	1	0	0	0	0	0	
KNN Classification over Test Set 1																															
Class 1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
Class 2	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 3	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	0	0	0	0	1	0	0	0	1	0	0	0	0	1	0	0
Class 4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 5	0	1	0	0	1	1	1	0	1	1	1	1	0	0	0	0	1	1	1	0	1	1	1	0	1	1	1	0	0	0	0

Table 3. Simulation 3.

Target of Test Set 1																															
Image	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
Class 1	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	
Class 2	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 3	0	0	0	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1
Class 4	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0
Class 5	1	1	1	0	0	1	0	0	0	1	0	0	0	0	0	1	1	1	0	1	1	1	0	1	1	1	1	0	0	0	0
NB Classification over Test Set 1																															
Class 1	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	
Class 2	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 3	0	0	0	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1
Class 4	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0	0
Class 5	1	1	1	0	0	1	0	0	0	1	0	0	0	0	0	1	1	1	0	1	1	1	0	1	1	1	1	0	0	0	0
KNN Classification over Test Set 1																															
Class 1	0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0	
Class 2	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Class 3	0	0	0	1	1	0	0	0	0	0	0	1	1	1	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	1
Class 4	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
Class 5	1	1	1	0	0	1	0	0	0	1	0	0	0	0	0	1	1	1	0	1	1	1	0	1	1	1	0	1	0	1	0

6 Conclusions

The NB and KNN-based model is capable of recognizing several maturity levels of orange images using their descriptors. For all tests the average of accuracy is over

90.00% without image preprocessing. This is an important issue because better descriptors values can be obtained using image preprocessing algorithms in order to improve the classification process.

The winner classes returned by NB and KNN methods are similar and close to correct ones. This means that it is possible that the classifiers detect human errors at the moment of building pattern set. Therefore, the mentioned methods can be used to refine the clustering process.

Both methods can be used at the same time and to test whether they return equal results, in which case, the new classified instance might be incorporated to training set to improve the classification.

The necessary computational resources are very small. For this reason it is possible to implement the proposed models and work with them in real time.

Future Work

Presently, the proposed model in this paper is being compared with Artificial Neural Network architectures in order to obtain best results on image classification.

In addition to the mentioned, Principal Component Analysis process of descriptors is being considered to reduce the problem dimensionality and to obtain the most representative data set descriptors.

Acknowledgment

This work has been supported by the project “Sistema Automático no Destructivo para la Clasificación de la Calidad de Citrus Regionales Basado en Visión por Computadora” Código: 25L/037 (National Technological University, Argentina).

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